

Lecture 4: Physical Layer

Wireless Communications

EC 441 - Introduction to Computer Networking
Boston University

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Learning Objectives

By the end of this lecture, you should be able to:

- Describe basic wireless transmitter and receiver architectures
- Calculate path loss using free-space and empirical models
- Perform wireless link budget calculations
- Explain how modulation enables channel selection and sharing
- Understand QAM and constellation diagrams
- Identify spectrum allocations for WiFi, cellular, and other services
- Analyze cellular frequency reuse patterns
- Understand interference and capacity limits in wireless networks

Why Wireless is Different

Key Differences from Wired:

- **Broadcast medium:** Everyone can hear your transmission
- **Path loss:** Signal strength decreases rapidly with distance
- **Interference:** Multiple transmitters share spectrum
- **Fading:** Signal varies due to multipath, obstacles
- **Regulation:** Spectrum is licensed or restricted
- **Power constraints:** Battery-powered devices

Benefits:

- **Mobility:** No physical connection needed
- **Deployment:** No cable installation
- **Flexibility:** Easy to add/remove devices

Basic Transmitter Architecture

[Data] → [Encoder] → [Modulator] → [Upconverter] → [PA] → [Antenna]

Key Components:

1. **Encoder:** Error correction (FEC), interleaving, scrambling
2. **Modulator:** Maps bits to symbols (I and Q signals)
3. **Upconverter:** Translates baseband to RF carrier

$$s(t) = I(t) \cos(2\pi f_c t) - Q(t) \sin(2\pi f_c t)$$

4. **Power Amplifier (PA):** Amplifies to transmit power
 - Tradeoff: Power vs. linearity vs. efficiency
 - Typical efficiency: 20-40%
5. **Antenna:** Converts electrical signal to EM waves

Basic Receiver Architecture

[Antenna] → [LNA] → [Downconverter] → [Demodulator] → [Decoder] → [Data]

Key Components:

1. Low-Noise Amplifier (LNA):

- **Most critical component!**
- Minimizes noise figure (typically 1-3 dB)
- High gain (15-25 dB) reduces impact of later stages (Friis formula)

2. Downconverter: Translates RF to baseband/IF

3. Demodulator: Extracts I and Q signals, symbol recovery

4. Decoder: Error correction decoding

Antenna Fundamentals

Isotropic Antenna (Theoretical):

- Radiates equally in all directions (sphere)
- Reference for antenna gain (0 dBi)
- Doesn't exist in practice!

Antenna Gain (G): Concentration of power in preferred direction

- **Omnidirectional** (dipole): ~ 2 dBi
- **Directional** (Yagi): 6-20 dBi
- **Dish** (satellite): 30-50 dBi

Effective Aperture:

$$A_e = \frac{G\lambda^2}{4\pi}$$

Reciprocity: Same gain for transmit and receive

Free Space Path Loss (FSPL)

Friis Transmission Equation:

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d} \right)^2$$

Path Loss in dB:

$$L_{\text{FSPL}}(\text{dB}) = 32.45 + 20 \log_{10}(f_{\text{MHz}}) + 20 \log_{10}(d_{\text{km}})$$

Key Insights:

- Distance: **20 dB per decade** ($10\times$ distance $\Rightarrow$ 20 dB loss)
- Frequency: 20 dB per decade ($10\times$ frequency $\Rightarrow$ 20 dB loss)
- Higher frequencies lose more power (smaller effective aperture)

Example: WiFi Link

Parameters:

- Frequency: 2.4 GHz
- Distance: 100 m = 0.1 km
- Transmit power: 20 dBm (100 mW)
- Antenna gains: $G_t = G_r = 2$ dBi

Calculation:

$$\begin{aligned} L_{\text{FSPL}} &= 32.45 + 20 \log_{10}(2400) + 20 \log_{10}(0.1) \\ &= 32.45 + 67.6 - 20 = 80 \text{ dB} \\ P_r &= P_t + G_t + G_r - L \\ &= 20 + 2 + 2 - 80 = -56 \text{ dBm} \end{aligned}$$

Well above typical WiFi sensitivity (-80 to -90 dBm) $\Rightarrow$ good link!

Empirical Path Loss Model:

$$L(d) = L_0 + 10n \log_{10} \left(\frac{d}{d_0} \right) + X_\sigma$$

where n = path loss exponent, X_σ = shadow fading

Environment	n	Loss/decade
Free space	2	20 dB
Urban cellular	2.7-3.5	27-35 dB
Indoor (LOS)	1.6-1.8	16-18 dB
Indoor (NLOS)	3-5	30-50 dB
Obstructed urban	4-6	40-60 dB

Key insight: Obstacles dramatically increase path loss!

Shadow Fading (Slow Fading):

- Random variations due to obstacles
- Log-normal distribution: $X_\sigma \sim \mathcal{N}(0, \sigma^2)$
- Typical $\sigma = 4 - 12$ dB
- Changes slowly (seconds to minutes)

Multipath Fading (Fast Fading):

- Multiple signal paths arrive at receiver
- Constructive and destructive interference
- Rayleigh fading (no LOS) or Rician fading (LOS + multipath)
- Changes rapidly (milliseconds)
- Can cause 20-30 dB signal variation!

Impact: Need 10-20 dB fading margin in link budget

Wireless Link Budget

Link Budget Equation:

$$P_r(\text{dBm}) = P_t(\text{dBm}) + G_t(\text{dBi}) + G_r(\text{dBi}) - L_{\text{path}}(\text{dB}) - L_{\text{misc}}(\text{dB})$$

Margin Calculation:

$$\text{Margin} = P_r - S_{\text{min}}$$

where S_{min} = receiver sensitivity

Adequate Margins:

- 10-15 dB: Acceptable for static links
- 20-30 dB: Needed for mobile/fading environments
- 40+ dB: Excellent, very reliable

Why Modulation?

Baseband Transmission Problems:

- Antenna size $\approx \lambda/4$ (at 1 kHz: 75 km antenna!)
- All users share same spectrum (can't separate)
- Propagation poor at low frequencies

Solution: Modulation

- Shift baseband signal to higher carrier frequency
- Different users/channels use different carriers
- Practical antenna sizes
- Frequency Division Multiple Access (FDMA)

Quadrature Amplitude Modulation (QAM)

Key Idea: Modulate both amplitude and phase

Transmitted Signal:

$$s(t) = I(t) \cos(2\pi f_c t) - Q(t) \sin(2\pi f_c t)$$

where:

- $I(t)$ = in-phase component
- $Q(t)$ = quadrature component
- These are **independent** data streams!

Complex Envelope:

$$z(t) = I(t) + jQ(t)$$

Visualized in constellation diagram (I-Q plane)

Constellation Diagrams

BPSK (Binary Phase Shift Keying):

- 2 symbols: ± 1
- 1 bit/symbol
- Minimum distance: $d = 2$

QPSK (Quadrature PSK):

- 4 symbols at $\pm 1 \pm j$
- 2 bits/symbol
- Distance: $d = \sqrt{2}$

16-QAM: 16 symbols in 4×4 grid, 4 bits/symbol

64-QAM, 256-QAM, 1024-QAM: Used in WiFi 6, 5G

- Higher spectral efficiency
- Require excellent SNR (>30 dB)

QAM Performance Requirements

Required SNR for $\text{BER} = 10^{-6}$:

Modulation	E_b/N_0 (dB)	Bits/symbol
BPSK	10.5	1
QPSK	10.5	2
16-QAM	14.5	4
64-QAM	18.5	6
256-QAM	23.5	8

Key Insight: Each doubling of M requires $\sim 3\text{-}4$ dB more SNR

Adaptive Modulation

Modern systems adapt modulation to channel quality

WiFi (802.11ac/ax) Rate Table:

MCS	Modulation	Code Rate	Rate (80 MHz)
0	BPSK	1/2	29 Mb/s
1	QPSK	1/2	58 Mb/s
3	16-QAM	1/2	117 Mb/s
5	16-QAM	3/4	175.5 Mb/s
7	64-QAM	5/6	263 Mb/s
9	256-QAM	3/4	351 Mb/s
11	1024-QAM	3/4	468 Mb/s

Far from AP: Low MCS (robust, slow) — **Near AP:** High MCS (fast, fragile)

Radio Frequency Spectrum

Band	Frequency	Wavelength	Use
VLF	3-30 kHz	100-10 km	Submarine comm
LF	30-300 kHz	10-1 km	Navigation
MF	300-3000 kHz	1 km-100 m	AM radio
HF	3-30 MHz	100-10 m	Shortwave
VHF	30-300 MHz	10-1 m	FM radio, TV
UHF	300-3000 MHz	1 m-10 cm	TV, cellular, WiFi
SHF	3-30 GHz	10-1 cm	Satellite, 5G
EHF	30-300 GHz	1 cm-1 mm	mmWave 5G

Cellular Spectrum (US)

Licensed Bands (exclusive use, FCC auctions):

4G LTE Frequencies:

- **Band 2 (PCS):** 1850-1910 MHz (UL), 1930-1990 MHz (DL)
- **Band 4 (AWS):** 1710-1755 MHz (UL), 2110-2155 MHz (DL)
- **Band 12 (700 MHz):** 699-716 MHz (UL), 729-746 MHz (DL)

5G Frequencies:

- **Low band (Sub-6 GHz):** 600-900 MHz (wide coverage)
- **Mid band (C-band):** 3.7-3.98 GHz (capacity + coverage)
- **High band (mmWave):** 24-47 GHz (ultra-capacity, short range)

FDD: Separate uplink/downlink frequencies

TDD: Same frequency, alternate in time

WiFi Spectrum (Unlicensed ISM Bands)

Anyone can use these bands (with power limits):

2.4 GHz Band (2.400-2.4835 GHz):

- Channels 1-11 (US), 20 MHz width
- Non-overlapping: Channels 1, 6, 11
- Crowded: Bluetooth, microwaves, cordless phones
- Good range (lower frequency)

5 GHz Band (5.150-5.825 GHz):

- Many channels, widths: 20, 40, 80, 160 MHz
- Less crowded, shorter range
- Higher speed (wider channels)

6 GHz Band (5.925-7.125 GHz): WiFi 6E

- 1200 MHz of spectrum! Very wide channels (320 MHz)

Other Wireless Systems

Bluetooth (2.4 GHz ISM):

- 79 channels, 1 MHz each
- Frequency hopping (1600 hops/sec)
- Short range (10-100 m), low power

GPS:

- L1: 1575.42 MHz, L2: 1227.60 MHz
- Receive-only, weak signal (-160 dBW!)

Satellite:

- C-band: 4-8 GHz, Ku-band: 12-18 GHz, Ka-band: 26-40 GHz

LoRaWAN (IoT):

- 915 MHz (US), ultra-long range (10+ km), very low data rate

Licensed	Unlicensed
Exclusive use	Shared
Expensive (auctions)	Free
Regulated power/quality	Best effort
Predictable performance	Interference possible
Example: Cellular	Example: WiFi, Bluetooth

Spectrum is incredibly valuable!

- Recent US C-band auction (2021): **\$81 billion**
- Limited resource (like real estate)
- Tradeoff: coverage (low freq) vs. capacity (high freq)

The Cellular Concept

Key Idea: Divide coverage area into cells

- Each cell has a base station
- **Reuse frequencies** in non-adjacent cells
- Increases total system capacity

Without Reuse:

- One transmitter covers entire city
- All users share single channel set
- Very limited capacity

With Reuse:

- Many transmitters (cell towers)
- Same frequencies used in distant cells
- **Capacity multiplied by reuse factor**

Hexagonal Cell Patterns

Why Hexagons?

- Approximate circular coverage
- Tile without gaps
- Geometric regularity

Cell Cluster: Group of N cells using different frequencies

Common Patterns:

- $N = 3$: Aggressive reuse $\Rightarrow$ more capacity, higher interference
- $N = 4$: Moderate reuse, balance capacity and quality
- $N = 7$: Conservative reuse $\Rightarrow$ lower interference, lower capacity

Used in early cellular (AMPS)

Reuse Distance

Distance to Nearest Co-Channel Cell:

$$D = R\sqrt{3N}$$

where R = cell radius, N = cluster size

Carrier-to-Interference Ratio (C/I):

Assuming 6 interfering cells in first tier:

$$\frac{C}{I} = \frac{(3N)^{n/2}}{6}$$

where n = path loss exponent

For $n = 4$ (typical urban):

- $N = 3$: $C/I = 18.6$ dB
- $N = 4$: $C/I = 24$ dB
- $N = 7$: $C/I = 31$ dB

Higher $N \Rightarrow$ Better C/I $\Rightarrow$ Lower capacity per cell

Modern Cellular: Reuse Factor = 1

Legacy Systems (2G GSM, 3G):

- Used $N = 3, 4$, or 7
- Different frequencies in adjacent cells
- Simple, but wastes spectrum

Modern Systems (4G LTE, 5G NR):

- **Reuse factor = 1** (all cells use all frequencies!)
- Advanced techniques:
 - Orthogonal Frequency Division Multiple Access (OFDMA)
 - Interference coordination between base stations
 - Beamforming (directional antennas)
 - Power control
 - Advanced receivers (interference cancellation)

Result: Much higher spectral efficiency!

Sectoring

Problem: Omnidirectional antennas waste power and create interference

Solution: Divide cell into sectors

- Typically **3 sectors per cell** (120° each)
- Directional antennas (17-21 dBi gain)
- Each sector like a separate cell

Benefits:

- 3× capacity (3 sectors)
- Lower interference (directional)
- Better link budget (antenna gain)

Towers Typically Have:

- 3 sectors (120° each)
- Multiple antennas per sector (MIMO)
- Multiple frequency bands

Key Takeaways

1 Wireless Architectures

- TX: encoder → modulator → upconverter → PA → antenna
- RX: antenna → LNA → downconverter → demodulator → decoder
- LNA is critical (first stage dominates noise figure)

2 Propagation

- Free space: 20 dB per decade of distance
- Real-world: path loss exponent $n = 2$ to 6
- Shadow fading: 4-12 dB variation
- Multipath fading: 20-30 dB variation (fast)

3 Link Budgets

- $P_r = P_t + G_t + G_r - L$
- Need 20-30 dB margin for fading

Key Takeaways (continued)

4 Modulation and QAM

- Modulation shifts baseband to carrier (channel selection)
- QAM: independent I and Q streams
- Higher order QAM $\Rightarrow$ better efficiency, needs higher SNR
- Adaptive modulation: change MCS based on channel

5 Spectrum Allocation

- Licensed: cellular (exclusive, expensive)
- Unlicensed: WiFi, Bluetooth (shared, free)
- 2.4 GHz: crowded, good range
- 5/6 GHz: less crowded, shorter range
- mmWave (24+ GHz): huge capacity, very short range

6 Cellular Reuse

- Cluster size N : tradeoff capacity vs. interference
- Modern systems: $N = 1$ with advanced interference management
- Sectoring: $3\times$ capacity with directional antennas

Connection to Networking

Physical layer constraints affect everything above:

- Variable data rates (adaptive modulation)
- Higher error rates than wired
- Shared medium (requires MAC protocol)
- Hidden/exposed terminal problems

WiFi design driven by physical layer:

- CSMA/CA (sense before transmit)
- RTS/CTS (solve hidden terminal)
- Channel selection (avoid interference)

Cellular optimized for mobility:

- Handoff between cells
- Power control (don't waste battery)
- Resource scheduling (fairness)

Data Link Layer:

- How to share the wireless channel (MAC protocols)
- Error detection and correction
- Frame formats
- WiFi (802.11) and Ethernet (802.3) in detail
- Medium access control
- Bridging and switching